

Table of Contents

File Operations	3
Summary	3
APT Package Management	4
Summary	4
C Compilation	5
Summary	5
CleanUp and Verification	6
Summary	6

Table of Figures

Figure 1: File Operations	3
Figure 2: Apt Package Management	4
Figure 3: C Compilation Process	5
Figure 4: Cleanup and verification	6

File Operations

```
kshitiz@kshitiz-VirtualBox:~$ mkdir new_linux
kshitiz@kshitiz-VirtualBox:~$ cd
kshitiz@kshitiz-VirtualBox:~$ cd new_linux
kshitiz@kshitiz-VirtualBox:~/new_linux$ nano greeting.txt
kshitiz@kshitiz-VirtualBox:~/new_linux$ cat greeting.txt
Hello !! I am Kshitiz Pokharel.
kshitiz@kshitiz-VirtualBox:~/new_linux$ cp greeting.txt backup.txt
kshitiz@kshitiz-VirtualBox:~/new_linux$ mv backup.txt old_greeting.txt
kshitiz@kshitiz-VirtualBox:~/new_linux$ rm old_greeting.txt
kshitiz@kshitiz-VirtualBox:~/new_linux$ mkdir temp_dir
kshitiz@kshitiz-VirtualBox:~/new_linux$ rmdir temp_dir
kshitiz@kshitiz-VirtualBox:~/new_linux$ ls -l greeting.txt
-rw-rw-r-- 1 kshitiz kshitiz 33 Sep  8 18:10 greeting.txt
kshitiz@kshitiz-VirtualBox:~/new_linux$
```

[Figure 1: File Operations](#)

Summary

This terminal session shows fundamental Linux file management. A new directory was created, a text file with a greeting was added, and the file was copied, renamed, and deleted. A temporary directory was also created and removed, followed by listing the details of the remaining file.

APT Package Management

```
kshitiz@kshitiz-VirtualBox:~/new_linux$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://np.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://np.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://np.archive.ubuntu.com/ubuntu resolute-backports InRelease
9 packages can be upgraded. Run 'apt list --upgradable' to see them.
kshitiz@kshitiz-VirtualBox:~/new_linux$ sudo apt install build-essential
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30      linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30-gener
  linux-headers-7.0.0-30-generic  linux-modules-7.0.0-30-generic
  linux-image-unsigned-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Processing triggers for libc-bin (2.41-0ubuntu1) ...
kshitiz@kshitiz-VirtualBox:~/new_linux$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

kshitiz@kshitiz-VirtualBox:~/new_linux$ apt search python3
alembic/resolute 1.18.4-1 all
  lightweight database migration tool for SQLAlchemy

antimony/resolute 0.9.3-2.1build2 amd64
  Computer-aided design CAD tool

astro-frameworks/resolute 5.0 all
  Astronomical data analysis frameworks

astro-python3/resolute 5.0 all
  Python 3 packages for astronomy
```

[Figure 2: Apt Package Management](#)

Summary

This terminal session highlights software setup tasks on Ubuntu. First, the GCC compiler version was checked, confirming installation. Then, a search for Python 3 packages was performed, showing several available options in the repository. Overall, it shows verifying development tools and exploring Python-related packages for system configuration.

C Compilation

```
kshitiz@kshitiz-VirtualBox:~/new_linux$ nano hello.c
kshitiz@kshitiz-VirtualBox:~/new_linux$ gcc hello.c -o hello_program
kshitiz@kshitiz-VirtualBox:~/new_linux$ ls -l hello_program
-rwxrwxr-x 1 kshitiz kshitiz 15992 Sep  8 18:38 hello_program
kshitiz@kshitiz-VirtualBox:~/new_linux$ ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 is 15
kshitiz@kshitiz-VirtualBox:~/new_linux$ gcc -Wall hello.c -o hello_program_warn
kshitiz@kshitiz-VirtualBox:~/new_linux$ ./hello_program_warn
bash: ./hello_program_warn: No such file or directory
kshitiz@kshitiz-VirtualBox:~/new_linux$ ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 is 15
kshitiz@kshitiz-VirtualBox:~/new_linux$ nano hello.c
kshitiz@kshitiz-VirtualBox:~/new_linux$ gcc -Wall hello.c -o hello_program
kshitiz@kshitiz-VirtualBox:~/new_linux$ ./hello_program
Hello, Linux Lab!
Sum of 100 and 5 is 105
kshitiz@kshitiz-VirtualBox:~/new_linux$
```

[*Figure 3: C Compilation Process*](#)

Summary

A file called `hello.c` was made and compiled using `gcc`. The output file `hello_program` was created and checked. When the program was run, it showed the message “Hello, Linux Lab!” and printed sums like $10+5=15$ and $100+5=105$. This shows the steps of writing, compiling, and running a small C program in Linux.

CleanUp and Verification

```
Sum of 100 and 5 is 105
kshitiz@kshitiz-VirtualBox:~/new_linux$ cd
kshitiz@kshitiz-VirtualBox:~$ ls -lh new_linux
total 72K
-rw-r--r-- 1 kshitiz kshitiz 24K Sep  8 18:28 exit
-rw-rw-r-- 1 kshitiz kshitiz  33 Sep  8 18:10 greeting.txt
-rw-rw-r-- 1 kshitiz kshitiz  34 Sep  8 18:10 greeting.txtty
-rw-rw-r-- 1 kshitiz kshitiz 158 Sep  8 18:41 hello.c
-rw-rw-r-- 1 kshitiz kshitiz 127 Sep  8 18:31 hello.cy
-rwxrwxr-x 1 kshitiz kshitiz 16K Sep  8 18:41 hello_program
-rwxrwxr-x 1 kshitiz kshitiz 16K Sep  8 18:39 hello_program_warn
kshitiz@kshitiz-VirtualBox:~$ cat new_linux/hello.c

#include <stdio.h>
int main () {
    int a = 100, b = 5;
    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d is %d\n", a , b,  a+b);
    return 0;
}

kshitiz@kshitiz-VirtualBox:~$ rm -rf new_linux
```

Figure 4: Cleanup and verification

Summary

The folder was checked and files were listed. The code in hello.c prints “Hello, Linux Lab!” and shows that $100+5=105$. After this, the folder new_linux was deleted. This shows writing, running, and cleaning a small C program in Linux.